

06

Vector Space

Vector Space

A natural extension of the Cartesian coordinate system

6.1 Vector Space



Core skills you'll master in this section:

- ▶ A vector space is built from vector addition and scalar multiplication, and satisfies 8 basic axioms.
- ▶ Linear combination: a weighted sum of vectors.
- ▶ Linear dependence: at least one vector in a set of vectors can be linearly represented by the others; otherwise, the set is linearly independent.
- ▶ Span: the space generated by linear combinations.
- ▶ Maximal linearly independent subset: the largest possible subset of pairwise linearly independent vectors chosen from a set of vectors.
- ▶ Rank: the number of vectors in a maximal linearly independent subset.
- ▶ Basis: a set of linearly independent vectors whose linear combinations can uniquely represent any vector in the vector space.
- ▶ Dimension of a vector space: the number of vectors in the space's basis.
- ▶ Equivalent statements: a square matrix has full rank, a square matrix is invertible, a square matrix's column vectors are linearly independent, a square matrix's row vectors are linearly independent, and its determinant is nonzero.
- ▶ Translating a vector space produces an affine space, and the origin is no longer part of it.

Earlier chapters of this book have already touched on various common concepts of vector spaces wherever the opportunity arose. This chapter is a dedicated treatment of vector spaces.

A vector space is a natural extension of the Cartesian coordinate system. This section uses examples from two- and three-dimensional rectangular coordinate systems to show how vectors form the “skeleton” of a space. The definition of a vector space relies on closure under addition and scalar multiplication, and satisfies a series of axioms, such as commutativity, associativity, and the existence of an identity and an inverse. These axioms ensure that a vector space has a stable algebraic structure, laying the foundation for linear algebra. This section also reviews the concepts of linear combination, linear dependence, and linear independence, and further introduces rank, the maximal linearly independent subset, basis, the dimension of a space, and affine space.

Starting from the Cartesian Coordinate System

A vector space is a natural extension of the Cartesian coordinate system.

Figure 1 gives the two- and three-dimensional rectangular coordinate systems; in vector-space terms, these are the most basic Euclidean vector spaces $\mathbb{R}^n$ ($n = 2, 3$).

⚠ Please note that in Figure 1's two subplots, the $\mathbf{e}_1$ and $\mathbf{e}_2$ dimensions (number of components) are not the same. In Figure 1 (a), $\mathbf{e}_1$ and $\mathbf{e}_2$ are two-dimensional; in Figure 1 (b), $\mathbf{e}_1$ and $\mathbf{e}_2$ are three-dimensional.

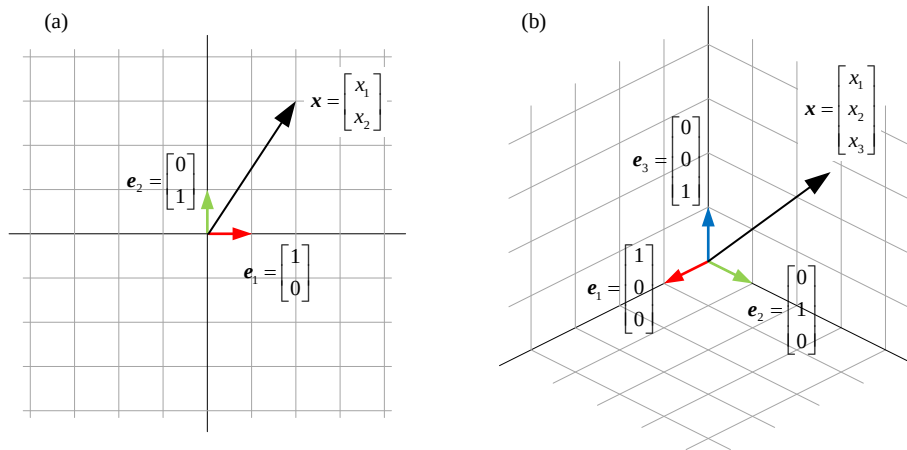


Figure 1. Two- and Three-Dimensional Rectangular Coordinate Systems

In the two vector spaces in Figure 1, we can perform a series of operations such as vector addition/subtraction and scalar multiplication.

$\mathbb{R}^2$, any vector $\mathbf{x}$ can be written as,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2$$

The equation above involves both scalar multiplication of vectors and vector addition.

As shown in Figure 1 (a), on $\mathbb{R}^2$, when x_1 and x_2 take arbitrary real values, the vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ covers the entire plane.

The two-dimensional column vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ prop up this plane like a “skeleton.” Clearly, $\mathbf{e}_1$ and $\mathbf{e}_2$ — these two skeleton vectors — are orthogonal.

Similarly, in $\mathbb{R}^3$, any vector $\mathbf{x}$ can be written as,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + x_3 \mathbf{e}_3$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

As shown in Figure 1 (b), on $\mathbb{R}^3$, when x_1 , x_2 , and x_3 take arbitrary real values, the three-dimensional column vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ prop up this space like a “skeleton.” Clearly, $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ — these three skeleton vectors — are pairwise orthogonal.

Vector Space

Next, let's look at the precise definition of a vector space.

Given a field F , a vector space V over F is a set. The set V is nonempty, and is closed under addition and scalar multiplication. This means that for every pair of elements $\mathbf{a}$ and $\mathbf{b}$ in V , there corresponds a unique element $\mathbf{a} + \mathbf{b}$ in V ; and for every element $\mathbf{a}$ in V and any scalar k , there corresponds a unique element $k\mathbf{a}$ in V .

If V , together with the addition and scalar multiplication operations above, satisfies the following axioms, then V is called a vector space.

⚠ Note that the definition of a vector space includes a series of axioms, whose purpose is to ensure that a vector space has a rigorous algebraic structure, thereby providing a solid foundation for the various operations and theories of linear algebra. There's no need to memorize these axioms by rote! Just treat them as a review of the common vector operations from Chapter 1 of this book.

The first 4 axioms are all related to vector addition and subtraction.

Axiom 1: Commutativity of Vector Addition

For any $\mathbf{a}$ and $\mathbf{b}$ in V , the following holds:

$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$$

As shown in Figure 2, in the RGB color space, combining red and blue produces magenta. Whether it's “red + blue” or “blue + red,” the final color is magenta:

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Red Blue Blue Red

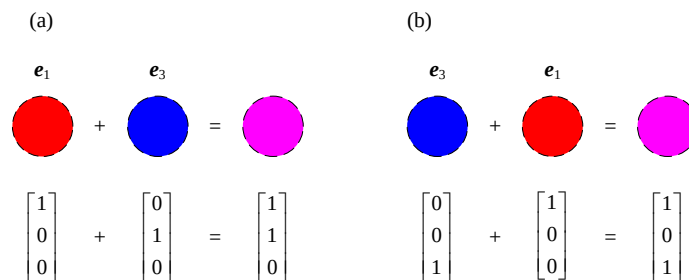


Figure 2. Commutativity of Vector Addition

Axiom 2: Associativity of Vector Addition

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For any $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ in V , the following holds:

$$(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$$

In the RGB color space, red and green first combine to produce yellow, and then adding blue gives white:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

If we first mix green and blue to get cyan, and then add red, the result is the same:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

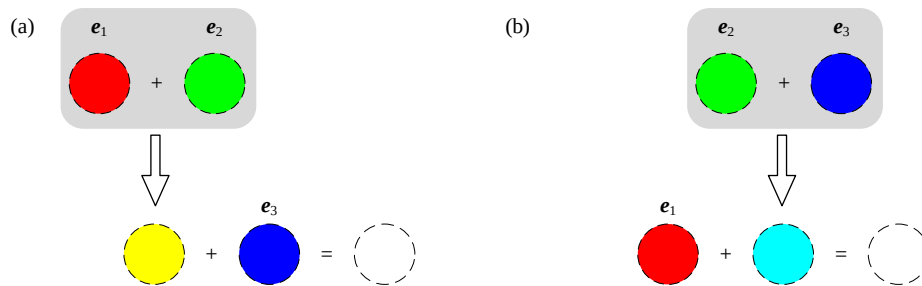


Figure 3. Associativity of Vector Addition

Axiom 3: Additive Identity of Vector Addition

There exists a zero vector $\mathbf{0}$ in V such that for any element $\mathbf{a}$ in V , the following holds:

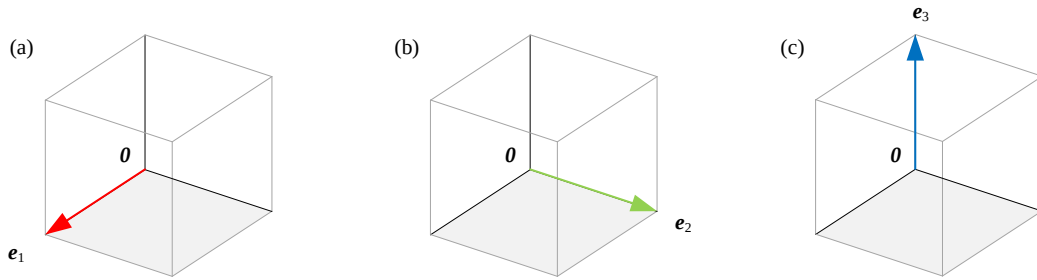
$$\mathbf{a} + \mathbf{0} = \mathbf{a}$$

This zero vector $\mathbf{0}$ is the additive identity of vector addition, also called the additive identity element.

In RGB color, the zero vector $\mathbf{0}$ can be thought of as black. For any color, for example, blue, adding black (no light) does not change the color

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

As shown in Figure 4, from a geometric perspective, adding a color vector to the zero vector $\mathbf{0}$ is equivalent to placing the color vector's starting point at the origin.

Figure 4. A Color Vector Added to the Zero Vector $\mathbf{0}$ **Axiom 4: Existence of an Additive Inverse**

For every element $\mathbf{a}$ in V , there is another element $-\mathbf{a}$ in V , satisfying:

$$\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$$

In RGB color, there's no such thing as a true negative color, because each RGB component's range is $[0, 1]$, and a negative color value has no practical meaning. However, we can think of the negative of a vector as removing color. For example, removing red light from red light itself gives black (the zero vector $\mathbf{0}$).

The next 4 axioms are related to scalar multiplication.

Axiom 5: Distributivity of Scalar Multiplication over Vector Addition

For any scalar k , the elements $\mathbf{a}$ and $\mathbf{b}$ in V satisfy:

$$k(\mathbf{a} + \mathbf{b}) = k\mathbf{a} + k\mathbf{b}$$

k can take any real value, and the RGB color space clearly does not satisfy this. This also shows that the RGB color space is not a true vector space in the strict sense; however, this doesn't stop us from using RGB to intuitively understand the key concepts of a vector space.

For RGB colors, if you increase the brightness of some “composite” color ($\times 0.5$), you can either mix the colors first and then adjust the brightness, or adjust the brightness of each color first and then add them – the result is the same.

As shown in Figure 5 (a), mix first, then adjust brightness

$$0.5 \times \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = 0.5 \times \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0 \\ 0.5 \end{bmatrix}$$

As shown in Figure 5 (b), adjust brightness separately, then mix

$$0.5 \times \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 0.5 \times \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0 \\ 0.5 \end{bmatrix}$$

Comparing Figure 5 (a) and Figure 5 (b), we find that these two paths ultimately produce the same color vector.

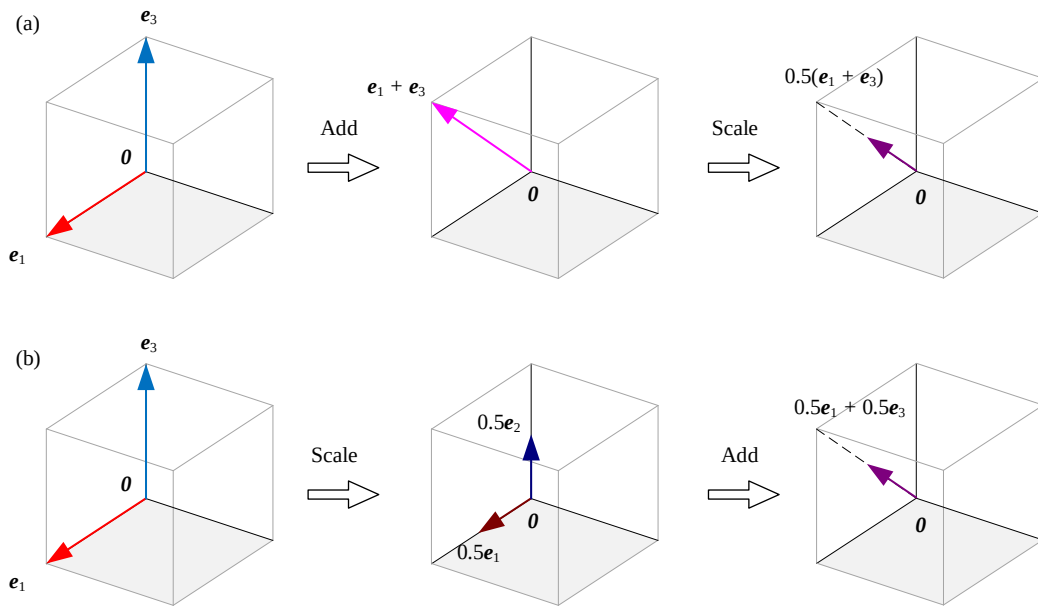


Figure 5. Distributivity of Scalar Multiplication over Vector Addition

Axiom 6: Distributivity of Scalar Multiplication over Scalar Addition

For any scalars k and t , and any element $\mathbf{a}$ in V , the following holds:

$$(k + t)\mathbf{a} = k\mathbf{a} + t\mathbf{a}$$

Suppose we have an orange color, and we want its intensity to become 0.5. We can adjust it in two steps, first by a factor of 0.3, then by a factor of 0.2; there are two ways to do this.

Method 1: First find the total brightness-change factor

$$(0.3 + 0.2) \times \begin{bmatrix} 1 \\ 0.5 \\ 0 \end{bmatrix} = 0.5 \times \begin{bmatrix} 1 \\ 0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.25 \\ 0 \end{bmatrix}$$

Method 2: Adjust separately, then add

$$0.3 \times \begin{bmatrix} 1 \\ 0.5 \\ 0 \end{bmatrix} + 0.2 \times \begin{bmatrix} 1 \\ 0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.3 \\ 0.15 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.2 \\ 0.1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.25 \\ 0 \end{bmatrix}$$

Axiom 7: Associativity of Scalar Multiplication

For any scalars k and t , and any element $\mathbf{a}$ in V , the following holds:

$$(kt)\mathbf{a} = k(t\mathbf{a})$$

For cyan, if we first scale down its intensity by a factor of 0.2, then by a factor of 0.5, there are two ways to do this.

Method 1: Directly compute the final brightness-change factor

$$(0.2 \times 0.5) \times \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = 0.1 \times \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.1 \\ 0.1 \end{bmatrix}$$

Method 2: Adjust step by step

$$0.2 \times \left(0.5 \times \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right) = 0.2 \times \begin{bmatrix} 0 \\ 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.1 \\ 0.1 \end{bmatrix}$$

Axiom 8: Identity Element of Scalar Multiplication

For any element $\mathbf{a}$ in V , the following holds:

$$1 \cdot \mathbf{a} = \mathbf{a}$$

No matter what color vector it is, multiplying by 1 does not change the color

$$1 \times \begin{bmatrix} 0.3 \\ 0.6 \\ 0.9 \end{bmatrix} = \begin{bmatrix} 0.3 \\ 0.6 \\ 0.9 \end{bmatrix}$$

These 8 axioms ensure that vector addition and scalar multiplication have well-behaved properties, allowing the basic concepts of linear algebra to be used in any vector space.

Linear Combination

I'm sure linear combinations are already familiar to you. Simply put, as shown in Figure 6, given a set of vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p$ and a set of scalars (weights) $k_1, k_2, \dots, k_p$, their linear combination is defined as:

$$\mathbf{b} = k_1 \mathbf{a}_1 + k_2 \mathbf{a}_2 + \dots + k_p \mathbf{a}_p$$

where $\mathbf{b}$ is the newly generated vector.

Of course, the equation above holds only if $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p$ all have the same dimension, that is, the same number of components.

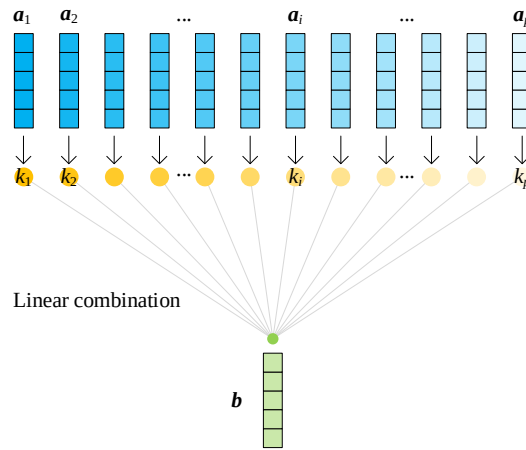


Figure 6. Linear Combination of Column Vectors

Linear Dependence and Linear Independence

Given a set of vectors $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p]$, if there exist $k_1, k_2, \dots, k_p$, not all zero, such that the following equation holds.

$$k_1 \mathbf{a}_1 + k_2 \mathbf{a}_2 + k_3 \mathbf{a}_3 + \dots + k_p \mathbf{a}_p = \mathbf{0}$$

then the set of vectors $\mathbf{A}$ is called linearly dependent (linear dependence; adjective form: linearly dependent); otherwise, $\mathbf{A}$ is linearly independent (linear independence; adjective form: linearly independent).

Let's unpack the equation above.

Linear independence means that no vector in a set of vectors can be expressed as a linear combination of the others, that is, these vectors are directionally independent of each other and cannot be replaced.

Linear dependence is the opposite: in a set of vectors, more than one vector can be expressed using the remaining vectors.

Let's go back to the RGB color space again, and explain linear dependence and linear independence from a geometric perspective.

Figure 7 (a) shows the vector group $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ which is clearly linearly dependent, since $\mathbf{e}_1 + \mathbf{e}_2$ can be expressed as the sum of $\mathbf{e}_1$ and $\mathbf{e}_2$. From another perspective, $\mathbf{e}_1 + \mathbf{e}_2$ lies in the plane spanned by $\mathbf{e}_1$ and $\mathbf{e}_2$.

Looking at it yet another way, $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ — deleting any one vector still spans the “red-green” plane. For example, deleting the yellow vector, the red and green vectors can span the “red-green” plane; deleting the green vector, the red and yellow vectors likewise span the “red-green” plane.

Similarly, in Figure 7 (b) and (c), the two vector groups are also linearly dependent; please analyze them on your own.

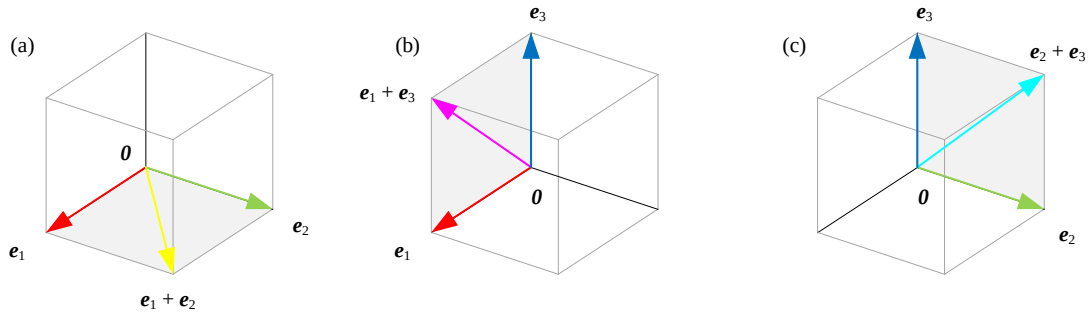


Figure 7. Linearly Dependent Color Vector Groups in the RGB Color Space, Groups of Three Vectors

Figure 8 shows three more vector groups that are also linearly dependent.

? Please think about which vectors, if deleted from each subplot of Figure 8, would make the remaining vectors linearly independent, and whether the choice of vector to delete is unique.

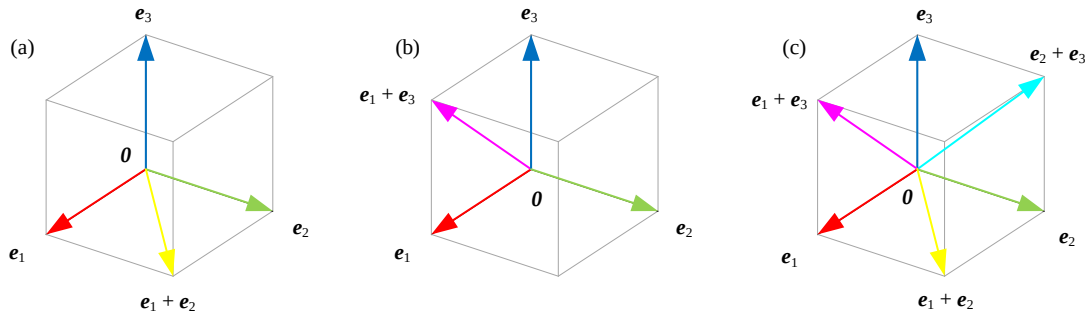


Figure 8. Linearly Dependent Color Vector Groups in the RGB Color Space, Groups of More Than Three Vectors

Figure 9 shows three more color vector groups, all linearly independent. For example, as shown in Figure 9 (a), $e_1 + e_2$, e_1 each make a special directional contribution that can't be expressed by the others – even though they aren't orthogonal. Please analyze Figure 9's remaining two subplots on your own.

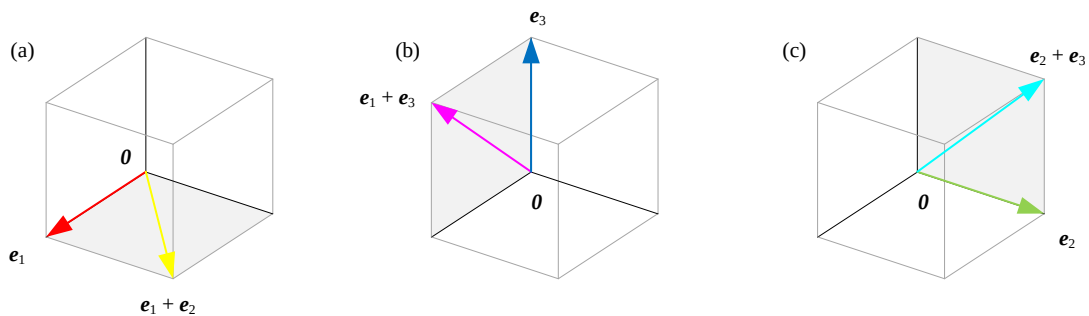


Figure 9. Linearly Independent Color Vector Groups in the RGB Color Space, Groups of Two Vectors

Figure 10 shows three vector groups that are all linearly independent.

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Figure 10 also tells us that a three-dimensional space needs at least three “skeleton” vectors to prop it up; the three “skeleton” vectors need not be pairwise orthogonal. This actually leads into the concepts of basis, orthogonal basis, and non-orthogonal basis, which this chapter will explain later.

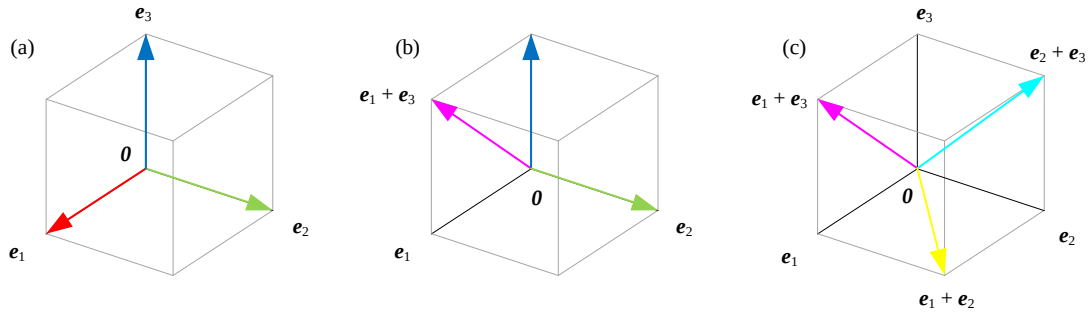


Figure 10. Linearly Independent Color Vector Groups in the RGB Color Space, Groups of Three Vectors

Span

$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p$ The set of all linear combinations of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p$ is called its span, denoted $\text{span}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p)$.

⚠ Note that for a span, the vector group can be either linearly dependent or linearly independent.

Let's again use the RGB color space as an example.

Figure 11 — the six vector groups shown all span the “red-green” plane.

Figure 11 (a), (b), (c) — these three vector groups are themselves all linearly independent.

And in Figure 11 (d), (e), (f), these three vector groups are themselves all linearly dependent.

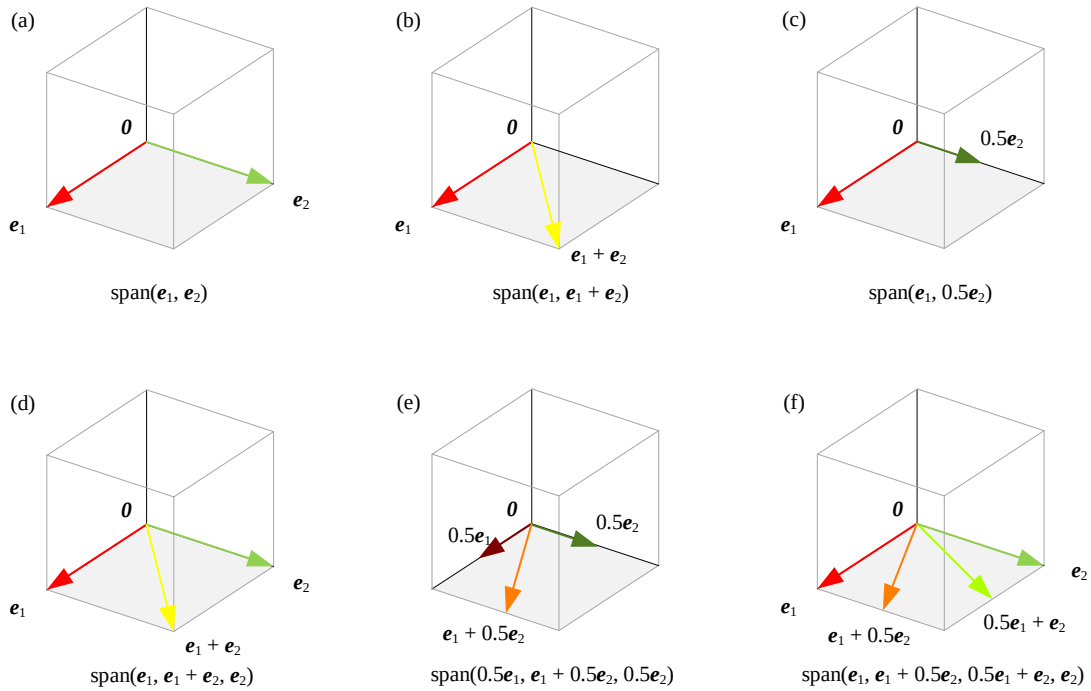


Figure 11. Different Vector Groups That Span the “Red-Green” Plane

Maximal Independent Subset and Rank

The column rank of the matrix A is the maximum number of linearly independent column vectors in A ; similarly, the row rank of A is the maximum number of linearly independent row vectors.

A matrix's column rank and row rank are always equal, so we simply call them the rank of the matrix A , denoted $\text{rank}(A)$.

If the column vectors in matrix A are linearly dependent, we can always find a redundant vector and remove it. Repeating this – continually removing redundant vectors until none remain – leaves a set of linearly independent vectors. This set is called a maximal linearly independent subset of A .

⚠ Note that the maximal linearly independent subset is not unique.

The number of elements in the maximal linearly independent subset equals the rank of the matrix.

Taking Figure 7 (a)'s vector group $[e_1, e_2, e_1 + e_2]$ as an example, we can remove redundant vectors to get the three maximal linearly independent subsets shown in Figure 12. Each maximal linearly independent subset has only two vectors.

That is, for the following matrix

$$[e_1 \quad e_2 \quad e_1 + e_2] = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

its column rank is 2; clearly, this matrix's row rank is also 2 (it has one row that is entirely zero).

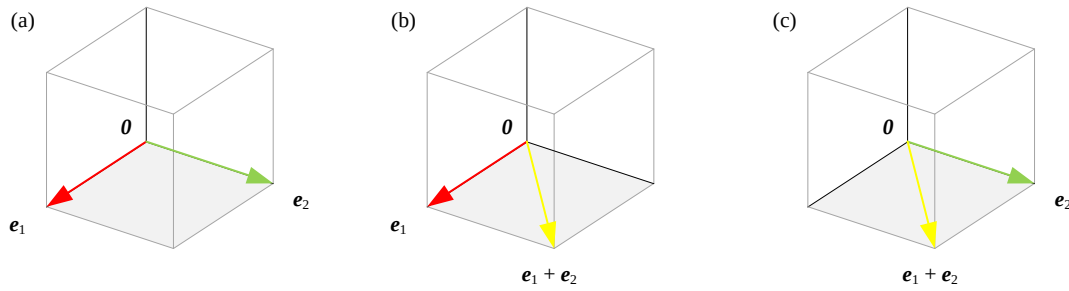


Figure 12. Removing Figure 7 (a)'s Redundant Vectors to Obtain a Maximal Linearly Independent Subset

Figure 8 (c) – removing several vectors gives the maximal linearly independent subsets shown in Figure 13. That is, the vector group in each of Figure 13's subplots can span the entire RGB color space.

? Please think about what other maximal linearly independent subsets exist for Figure 8 (c). Please draw them out.

In particular, if matrix A has p columns, then when $\text{rank}(A) = p$, matrix A has full column rank, and the column vectors $a_1, a_2, \dots, a_p$ are linearly independent.

! Please note that a square matrix $A_{p \times p}$ is invertible only when it has full rank, that is, $\text{rank}(A) = p$, then A is invertible.

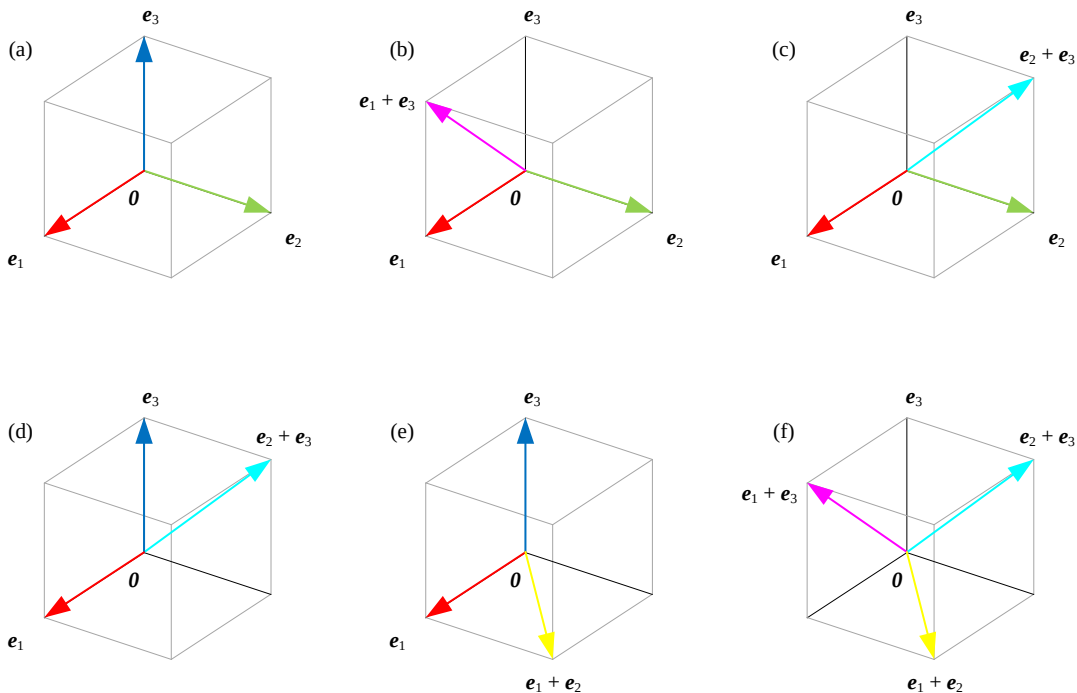


Figure 13. Removing Figure 8 (c)'s Redundant Vectors to Obtain a Maximal Linearly Independent Subset

Code 1 computes the rank of a matrix. **a** Defines three column vectors (as 2D arrays).

b Using the `np.column_stack()` function, `e_1`, `e_2`, and `e_3` are stacked as column vectors to form a matrix `A`. Then, `np.linalg.matrix_rank()` is used to compute the rank of matrix `A`.

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Code 1. Computing Rank |  Bk1_Ch06_01_01.ipynb

```

## 初始化
import numpy as np

## 定义列向量
a
e_1 = np.array([[1],[0],[0]])
e_2 = np.array([[0],[1],[0]])
e_3 = np.array([[0],[0],[1]])

## 计算秩
b
A = np.column_stack([e_1, e_2, e_3])
np.linalg.matrix_rank(A)

B = np.column_stack([e_1, e_2, e_1 + e_2])
np.linalg.matrix_rank(B)

C = np.column_stack([e_1, e_2])
np.linalg.matrix_rank(C)

D = np.column_stack([e_1, -e_1, e_1/2])
np.linalg.matrix_rank(D)

```

Basis and Basis Vectors

A basis (vector basis, or basis) is a set of linearly independent vectors in a vector space that can “prop up” the entire space.

A vector space V 's basis vectors are linearly independent vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p$ in V that span the vector space V , that is, $V = \text{span}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p)$.

And $[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p]$ is called a basis of V . Every vector in the vector space V can be uniquely expressed as a linear combination of the basis vectors in the basis $[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p]$.

Put simply, a basis is like the longitude and latitude on a map — it provides a way to pinpoint location. Once you have longitude and latitude, any point on the ground has a unique coordinate.

 Note that a span does not require the vector group to be linearly independent; however, the vector group that forms a basis must be linearly independent.

$[\mathbf{e}_1, \mathbf{e}_2]$ is the plane $\mathbb{R}^2$'s “most natural” basis; on the plane $\mathbb{R}^2$, every vector $\mathbf{x}$ can be uniquely expressed as $x_1\mathbf{e}_1 + x_2\mathbf{e}_2$. And (x_1, x_2) is $\mathbf{x}$'s coordinate in the basis $[\mathbf{e}_1, \mathbf{e}_2]$.

 Note the difference between $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $[\mathbf{e}_1, \mathbf{e}_2]$. This book uses $[\mathbf{e}_1, \mathbf{e}_2]$ to denote an ordered basis, that is, the basis elements are arranged in the order “first $\mathbf{e}_1$, then $\mathbf{e}_2$.” Whereas $\{\mathbf{e}_1, \mathbf{e}_2\}$ denotes a set, in which the basis vectors have no order. In addition, the ordered basis $[\mathbf{e}_1, \mathbf{e}_2]$ constructs the identity matrix. Unless stated otherwise, bases in this book are always ordered bases.

The basis $[\mathbf{e}_1, \mathbf{e}_2]$ is exactly the 2×2 identity matrix

$$[\mathbf{e}_1 \ \mathbf{e}_2] = \begin{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}_{2 \times 2}$$

And the basis $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ is exactly the 3×3 identity matrix

$$[\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3] = \begin{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}_{3 \times 3}$$

This is why we call $[\mathbf{e}_1, \mathbf{e}_2]$ and $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ the standard basis (standard basis, natural basis) — they are the most “natural” skeleton for spanning $\mathbb{R}^2$ and $\mathbb{R}^3$. However, there are infinitely many bases that span $\mathbb{R}^2$ and $\mathbb{R}^3$.

Dimension

The dimension of a vector space is the number of basis vectors in its basis, representing the “size” of the space; this book uses the notation $\dim()$ for dimension.

Figure 1 (a): $\mathbb{R}^2 = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$, that is, $\mathbb{R}^2$'s dimension $\dim(\mathbb{R}^2) = 2$, and $[\mathbf{e}_1, \mathbf{e}_2]$'s rank is also 2.

Figure 1 (b): $\mathbb{R}^3 = \text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$, that is, $\mathbb{R}^3$'s dimension $\dim(\mathbb{R}^3) = 3$, and $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$'s rank is 3.

Next, to understand the concept of dimension, let's look at a few more examples.

Figure 14 shows six vector spaces of dimension 1. From a geometric perspective, these vector spaces are all lines. Please note in particular that these lines all pass through the origin $\mathbf{0}$. That is, $\mathbf{0}$ lies in each of these vector spaces.

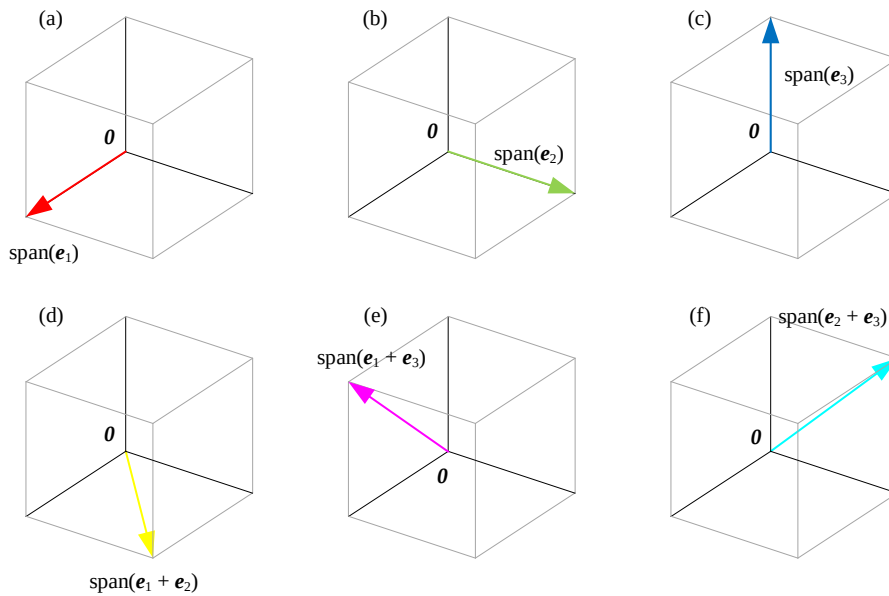


Figure 14. Vector Spaces of Dimension 1

Figure 15 shows dimension-2 vector spaces spanned by linearly independent vectors. That is, in each subplot of Figure 15, the two vectors are each the space's basis vectors. Once again, the basis vectors in a basis must be linearly independent.

From a set-theoretic perspective, $\text{span}(\mathbf{e}_1) \subset \text{span}(\mathbf{e}_1, \mathbf{e}_2)$, $\text{span}(\mathbf{e}_2) \subset \text{span}(\mathbf{e}_1, \mathbf{e}_2)$.

That is, $\text{span}(\mathbf{e}_1)$ is a subspace of $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$. Simply put, a subspace of a vector space is a nonempty subset of that vector space that is closed under vector addition and scalar multiplication, and contains the zero vector.

⚠ Note that $\mathbb{R}^2$ is not a subspace of $\mathbb{R}^3$; this is because the $\mathbb{R}^2$ and $\mathbf{e}_1$ that make up $\mathbf{e}_2$ are two-dimensional column vectors, that is, $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. While the $\mathbb{R}^3$, $\mathbf{e}_1$, and $\mathbf{e}_2$ that make up $\mathbf{e}_3$ are three-dimensional column vectors, that is, $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. The zero vector $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is in $\mathbb{R}^2$, but not in $\mathbb{R}^3$.

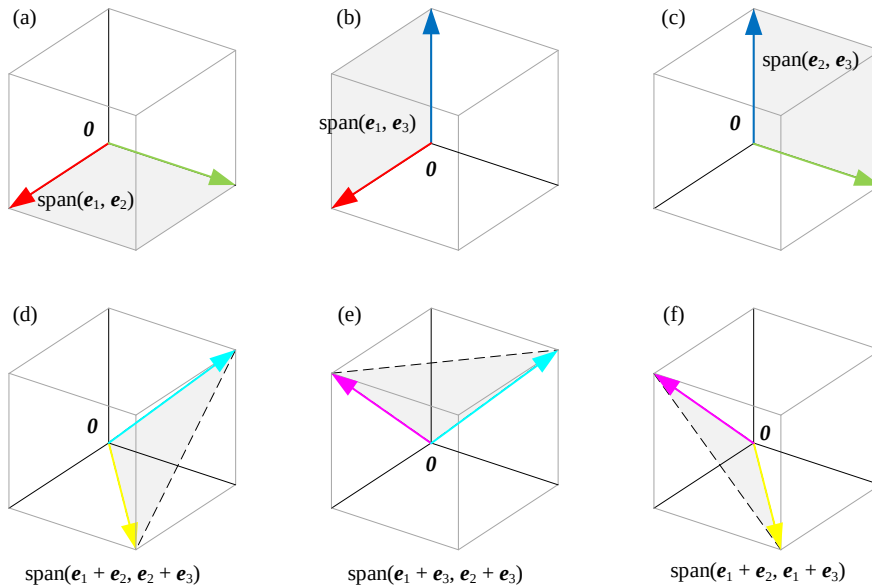


Figure 15. Vector Spaces of Dimension 2, Where the Spanning Basis Vectors Are Linearly Independent

Figure 16 shows dimension-2 spaces spanned by linearly dependent vectors.

For example, $\text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2)$ is a space of dimension 2. Clearly the vectors in $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ are linearly dependent, so $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ cannot be called a basis. Further analysis shows that $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ has rank 2.

The basis vectors in a basis must be linearly independent. After removing redundant vectors, $[\mathbf{e}_1, \mathbf{e}_2]$ and $[\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2]$ and $[\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ – any one of these three sets has linearly independent vectors, so all three can be chosen as a basis for the space $\text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2)$.

The difference is that in $[\mathbf{e}_1, \mathbf{e}_2]$, the basis vectors are orthogonal, but the vectors in the two bases $[\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2]$ and $[\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]$ are not orthogonal. That is, the basis vectors that make up a vector space can be either orthogonal or non-orthogonal – this is what we'll explore next.

I'm sure it's clear by now that the vectors within a basis must be linearly independent, while the vectors used in $\text{span}()$ to span a space can be linearly dependent. For example, $\text{span}(\mathbf{e}_1, \mathbf{e}_2) = \text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2) = \text{span}(\mathbf{e}_1, \mathbf{e}_2, 2\mathbf{e}_1, 2\mathbf{e}_1 + \mathbf{e}_2)$. In the basis $[\mathbf{e}_1, \mathbf{e}_2]$, every point has a unique coordinate. But in $\text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2)$, a point's coordinate is not unique.

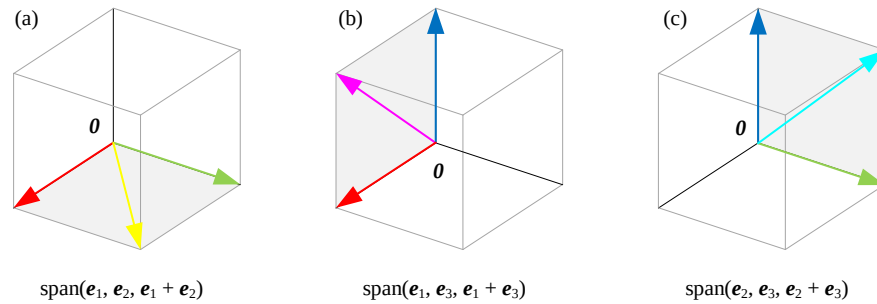


Figure 16. Vector Spaces of Dimension 2, Where the Spanning Vectors Are Linearly Dependent

Figure 17 shows dimension-3 spaces spanned by linearly independent vectors. Each vector group is a basis of that vector space.

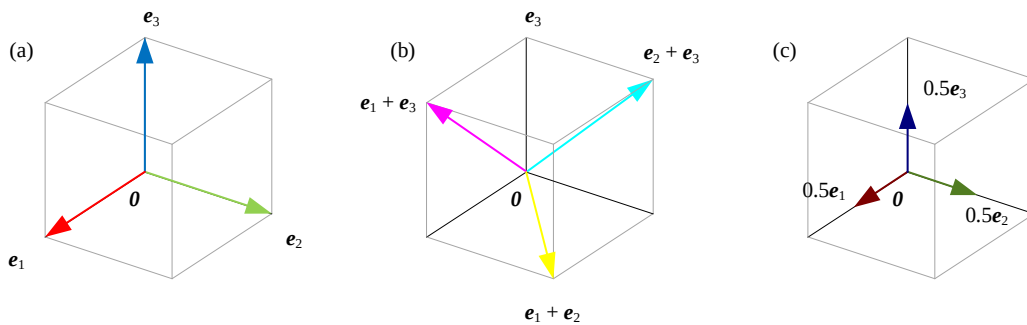


Figure 17. Vector Spaces of Dimension 3

Affine Space

The fact that a space “passes through the origin” is extremely important for a vector space. Figure 14 shows several one-dimensional spaces (lines) that clearly pass through the origin; that is, the origin $\mathbf{0}$ lies in the vector space. From a geometric perspective, Figure 15 and Figure 16 show dimension-2 spaces that are planes, and these planes all pass through the origin. The origin $\mathbf{0}$ also lies in Figure 17 the dimension-3 space shown.

The space obtained by translating a vector space is called an affine space, such as the three examples shown in Figure 18. The three affine spaces shown in Figure 18 clearly do not pass through the origin. This book will introduce affine transformations in detail later.

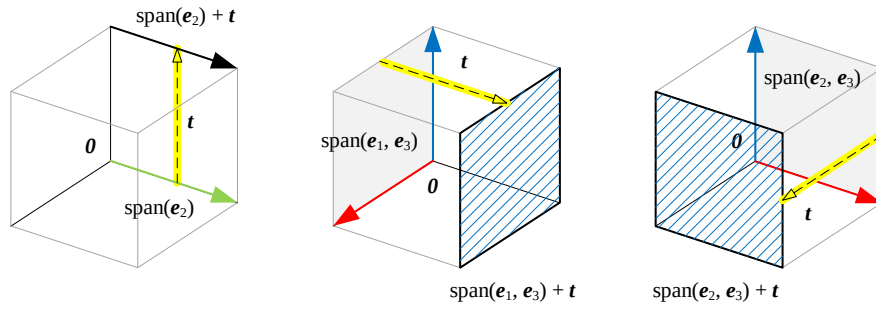


Figure 18. Translating a Vector Space to Obtain an Affine Space



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Which of the following are not a basis?

- ▶ $[e_1, e_2]$
- ▶ $[e_1, e_2, e_1 + e_2]$
- ▶ $[e_1, e_2, e_3]$
- ▶ $[e_2, e_3]$
- ▶ $[e_1, e_2, e_3, e_1 + e_2]$
- ▶ $[0.5e_1, e_2, 0.5e_3]$

Q2. For the vector groups in Q1 that are not a basis, which vector(s) should be removed from each to form a basis?

Q3. How are a square matrix's column vectors being linearly dependent or linearly independent related to the matrix's invertibility?

Q4. How is a square matrix having full column rank or not full column rank related to the matrix's invertibility?

Q5. How are a square matrix's column vectors being linearly dependent or linearly independent related to whether its determinant is 0?